

Amelia Eckard

Charlotte, NC | ameliaeckard.com | github.com/ameliaeckard | linkedin.com/in/ameliaeckard

Education

University of North Carolina at Charlotte Charlotte, NC
M.S. in Artificial Intelligence / Early Entry Program Expected May 2028

University of North Carolina at Charlotte Charlotte, NC
B.S. in Computer Science / Concentration: AI, Robotics, & Gaming Expected May 2027

Experience

Research Intern Remote
Stevens Institute of Technology, Dr. Jina Huh-Yoo June 2026 – Present

- Engineered *rq-system*/LEAF, an AI-assisted literature-analysis pipeline using PubMed E-utilities, CrossRef, and LLM APIs to screen, classify, and extract structured evidence from 345 papers; processed 256 included, 80 excluded, and 9 review-flagged papers.
- Implemented dataset tagging, a seven-category research-question taxonomy, explicit screening/classification/extraction gates, and gold-sample validation using precision, recall, F1, accuracy, and confusion matrices.

Undergraduate Researcher Charlotte, NC
UNC Charlotte, Dr. Todd Dobbs August 2025 – May 2026

- Developed an Apple Vision Pro indoor-navigation prototype for visually impaired users using Swift, ARKit, RealityKit, ObjectTrackingProvider, Core ML, HRTF spatial audio, and distance-based pitch cues.
- Investigated object tracking, on-device recognition, and spatial-audio guidance as complementary modalities for accessible indoor navigation. Designed evaluation protocols for object identification, tracking reliability, navigation performance, and spatial-audio feedback workflows.

Lead Instructional Assistant, ITSC 1213 Charlotte, NC
UNC Charlotte College of Computing and Informatics August 2026 – Present

- Serve as lead instructional assistant for an asynchronous Python programming course, coordinating grading, rubrics, office hours, student support, and course logistics with the instructor.
- Build and debug Python autograders and test suites for programming assignments, emphasizing executable behavior, edge cases, and clear feedback for students. Review submission patterns and course analytics to identify recurring implementation errors and improve assignment guidance, testing instructions, and pre-submission checks.

Instructional Assistant, ITSC 3160 Charlotte, NC
UNC Charlotte College of Computing and Informatics August 2026 – Present

- Support database-design students through grading, office hours, and targeted assistance with relational modeling, normalization, SQL, and semester project requirements.
- Provide debugging and design feedback that connects conceptual database models to working queries and implementation decisions.

Instructional Assistant, ITSC 1212 Charlotte, NC
UNC Charlotte College of Computing and Informatics June 2026 – August 2026

- Led introductory programming lab support covering control flow, functions, data structures, debugging, and foundational problem-solving.
- Reviewed student code, graded assignments, and provided individualized explanations to help students translate programming concepts into working solutions.

Peer Mentor Charlotte, NC
UNC Charlotte August 2025 – Present

- Support first-year students through individual check-ins, group discussions, events, study-strategy guidance, and navigation of university resources.

Projects

CNN Scene Classification | *Python, PyTorch, torchvision, CUDA* 2026

- Trained and evaluated convolutional neural networks for 16-class scene recognition on 2,400 images.
- Improved validation accuracy from 42.25% to 51.46% through image augmentation while comparing preprocessing choices, training behavior, and model performance.

Emergency Fund Predictor | *Python, scikit-learn, pandas, SHAP* 2025

- Built an 11-feature machine-learning pipeline using 12,295 Federal Reserve SHED respondents enriched with BEA indicators to study emergency-fund vulnerability.
- Compared three classifiers and applied SHAP explainability to identify the strongest drivers of financial fragility and model predictions.

- RoomCode** | *Python, Flask, JavaScript* 2026 – Present
- Built a multi-session collaborative coding environment with real-time synchronization, host-controlled execution, and isolated file trees.
 - Deployed the platform across 20+ lab sessions with concurrent users, supporting collaborative programming and classroom workflows.
- Scout Opportunity Notifier** | *Python, Discord API, SQLite, Railway* 2026 – Present
- Built and deployed an opt-in internship and hackathon aggregator with automated ingestion, deduplication, persistent subscriber preferences, and restart-safe SQLite storage.
 - Implemented personalized daily/weekly Discord digests, category-specific subscriptions, preference controls, unsubscribe flows, paginated listings, administrative broadcasts, and scheduled delivery windows.
- Hera's Garden** | *Node.js, Express, MariaDB, Discord.js, Java, Paper API, Railway* 2024 – Present
- Built and maintain a persistent creative-community platform spanning a full-stack website, Discord automation, and a custom Minecraft server ecosystem with scheduled releases, rank-gated access, OAuth2 administration, moderation, and persistent progression systems.
 - Designed Java/Paper systems for claims, territories, governments, housing, shops, mail, events, NPC societies, and economy features while integrating identity and progression across web, Discord, and game services.
- Iris** | *Electron, Node.js, Discord.js, SQLite, sqlite-vec, Ollama, MariaDB* 2025 – Present
- Developed a local-first personal-memory and community-automation system combining desktop tooling, persistent memory, local language models, and Discord operations.
 - Built verification, moderation, private-channel, XP/ranking, and notification workflows using SQLite/sqlite-vec and Ollama-based local retrieval and model components.
- Iris Calliope** | *Python, discord.py, SQLite, Fernet, LLMs, Railway* 2026 – Present
- Built a privacy-conscious Discord roleplay chronicle system that summarizes approved bot-generated activity without publicly ranking or profiling users.
 - Implemented encrypted storage, configurable retention, weekly summaries, channel watchlists, character lookup, and deletion of short-lived raw records while preserving higher-level continuity summaries.
- Vision Lab: ASCII Reconstruction** | *JavaScript, HTML/CSS, Image Processing* 2026 – Present
- Developed an image-to-ASCII reconstruction experiment for a web-based computer-vision lab connected to amelieackard.com.
 - Designed reconstruction logic around luminance mapping, local contrast, detail preservation, and edge-aware character selection to retain recognizable image structure.
- Resolve – LPL Financial University Hackathon** | *AI/ML, Workflow Intelligence, FinTech* 2026
- Developing a proactive exception-intelligence layer that predicts likely blockers before financial-service requests visibly stall.
 - Designing case-history and document understanding, workflow-state comparison, anomaly and stall-risk detection, causal explanations, and next-best-action recommendations.

Technical Skills

Languages Python, Java, JavaScript, Swift, SQL, C++, HTML/CSS

AI/Data PyTorch, torchvision, scikit-learn, NumPy, pandas, OpenCV, SHAP, Core ML, Ollama

Frameworks Flask, Node.js, Express, Discord.js, ARKit, RealityKit

Infrastructure/APIs SQLite, MariaDB, Git, Docker, Railway, CUDA, PubMed E-utilities, CrossRef, Perplexity API

Certifications & Involvement

Formal Committee Head, Alpha Omega Epsilon – Beta Kappa August 2025 – May 2026

- Coordinated formal-event planning, logistics, communication, and committee execution for the professional and social sorority.

Volunteer Judge & Field Reset, FIRST Robotics 2025 – Present

- Support robotics competitions through judging and field operations, helping maintain organized and fair event execution.

Kode With Klossy × Goldman Sachs Machine Learning Challenge December 2025

UR2PhD Training Course, Computing Research Association December 2025

Social and Behavioral Research, CITI Program September 2025